Contents

1 .Abstract..... 3

2. Introduction..... 3

3. Research Methodology:..... 3

Value Stream Mapping..... 4

Process Cycle Efficiency (PCE)..... 4

Interrelations Between 2 Tools:..... 5

Takt Time..... 5

....4.Data Analysis:..... **5**

5.Discussion:..... .8

6.Conclusion:..... 8

A study to reduce the takt time of a bakery factory by using lean tools

1 .Abstract

This study was carried out in a Bread production floor of “SunBest” Bakery factory in Sylhet, Bangladesh. The basic target of this study was to reduce the lead time of the observed Bread production floor using lean tools. The lean tools like Value Stream Map (VSM), Process Cycle Efficiency (PCE) were used in the process to reduce the lead time. Lead time is the summation of value added time(VD) and (NVD) time. It is an important metric for bakery factories, as it can impact customer satisfaction and profitability. We know that longer lead times are a challenge for a business for a business. It was observed that a number of causes were responsible for long lead time. In this study we have investigated the use of lean tools to reduce the lead times in order to improve the quality of the process.

2. Introduction

World is becoming more competitive for manufacturers as companies need to make their customers happy, keep their product quality high, control their costs and stay competitive. They need to design the process to satisfy the customer on the other hand they need to keep their costs and prices as low as possible to be able to compete with others by keeping their profitability. In order to achieve this, a company should have a very good control on its production system. A company should have a strong vision and clear idea of its future goals to establish the most effective production system. The application of lean tools can be well-defined strategy in these purposes. Many companies are using lean tools to get better at making things. These tools help cut down waste and also eliminate defects in manufacturing process which can lead to reduction in lead time, improved quality and reduced the overall cost.

However, this study was conducted in a Bread production floor of “SunBest” Bakery Company in Sylhet, Bangladesh. The stated bakery company was endeavored to reduce its long lead time, but they were not able to reduce it even though by taking big capital investment. In this report we have implemented the lean tools strategy to reduce the unwanted long lead time. It was also attempted to reduce the Takt time and improve the Process Cycle Efficiency (PCE) along with the reduced long lead time.

There are some basic steps that are followed in bread production process of this company. They are:

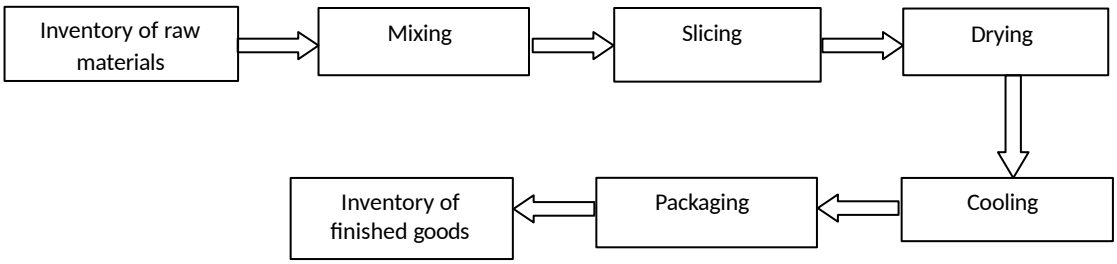


Figure1: Basic stages of Bread production

3. Research Methodology:

In this study, the "SunBest" Bakery factory was the population and its Bread production floor was the sample. The primary data were collected and documented by the direct observation of set up time, machine function, material and labor flow at each and every processing stages of the production line. The observed time was recorded by a stopwatch. The secondary data were collected through the internet, books, journals, related studies and other sources of information. To find out and to eliminate the manufacturing wastes and unwanted non value adding activities lean tools like VSM, PCE were used.

Value Stream Mapping:

Value Stream Mapping (VSM) is a visual tool used to identify and eliminate waste in a process. It maps out the flow of materials and information from the supplier to the customer, identifying all of the steps in the process, as well as the time and inventory associated with each step. VSM can be used to improve any process, from manufacturing to service delivery.

The VSM shows that the following steps are involved in the bread baking process:

1. Receive raw materials from supplier
2. Store raw materials in warehouse
3. Move raw materials to production line
4. Mix ingredients
5. Slicing
6. Drying
7. Cooling
8. Package bread
9. Store finished goods in warehouse
10. Ship finished goods to customer

Application: In a bread bakery, a Value Stream Map can be created to visually represent the entire process of making and delivering bread products to customers. This would include steps like mixing, proofing, baking, packaging, and distribution.

Process Cycle Efficiency (PCE):

- **Definition**: Process Cycle Efficiency is a metric used to assess the efficiency of a specific process within an organization. It measures the ratio of value-added time (the time spent on activities that add value) to the total cycle time (the time it takes for a product or service to move through the process from start to finish).

- **Formula:** PCE is typically calculated using the following formula:

  PCE = (Value-Added Time) / (lead time)

  Where, lead time = VD time +NVD time

- **Purpose:** PCE is used to identify and eliminate waste in a specific process. A higher PCE indicates a more efficient process with less non-value-added time or waste.
  A high PCE means that most of the time spent in the process is adding value to the final product or service. A low PCE indicates that a significant portion of the process time is non-value-added and should be reduced or eliminated to improve overall efficiency.
- **Application:** By calculating the PCE for each major process step, bakery managers can pinpoint which stages have the most non-value-added time. For example, if the PCE for baking is low due to long wait times or irregular heating, it suggests an area for improvement.

Interrelations Between 2 Tools:

- After creating a Value Stream Map, bakery managers can use the insights gained to prioritize which processes or areas need further analysis. For instance, if VSM shows that the packaging process is a bottleneck, PCE analysis can be conducted specifically for that step.
- PCE and VSM can be used in tandem to target improvements in processes with low efficiency. By comparing the PCE of different stages in the value stream, bakery managers can allocate resources to the most critical areas.

Takt Time

Takt time can be defined as the time required producing one unit of daily salable quantity. The purpose of takt time is to precisely match production with demand. It provides the heartbeat of a lean production system. At first, Takt time was used as a production management tool in the German aircraft industry in the 1930s.Takt time can be first determined with the formula:

$$T = T_a / T_d$$

Where

T = Takt time, e.g. [work time between two consecutive units]

T_a = Net time available to work, e.g. [work time per period]

T_d = Time demand (customer demand), e.g. [units required per period]

Net available time is the amount of time available for work to be done. This excludes break times and any expected stoppage time (for example scheduled maintenance, team briefings, lunch time etc.).

4.Data Analysis:

Required data for present VSM

Processing stage	VD time (sec)	NVD time (sec)	cycle time	mean downtime	Up time	change over time	Number of labor
Receiving inventory	0	12600	0	0		0	3
Mixing	558	787	590	107	81.864407	15	4
Slicing	1024	558	1110	278	74.954955	10	5
Drying	1395	1098	1475	398	73.016949	15	6
Cooling	1462	1348	1600	688	57	10	5
Packing	1180	433	1310	433	66.946565	10	6
Shipping inventory	0	11600	0	0		0	4
Total	5619	28424				60	

Present PCE of Bread production floor

At present state, VD and NVD time was calculated consecuitively 4181 sec and 28424 sec, so the lead time was 32605 sec. As earlier it was mentioned that -
PCE = (Value Added time/Lead time)*100%

PCE = 12.82318663

Present Takt time of Bread production floor

Working shift per day	= 2 shift
One shift time	32605 sec
Two shift time	65210 sec
Total change over time	= 1 hour or 3600 sec
Launch time	= 30 minutes or 1800 sec
Net available time per day	= Two shift time - (Total change over time + Launch time) = 65210sec -5400 sec = 59810 sec
Daily saleable demand	= 300 kg or 3000 bread

Takt time = 59810/3000 sec/bread
= 19.94 sec/bread

It seems that at present state, the PCE was 12.82%. But according to Zhen a productive floor PCE is need to be more than 25% to be globally competitive. In this study it was endeavored to increase the present PCE by using lean tools.

Required data for Future VSM

Processing stage	VD time (sec)	NVD time (sec)	cycle time	mean downtime	Up time	change over time	Number of labor
Receiving inventory	0	6300	0	0		0	3
Mixing	558	236	590	32	94.576271	15	4
Slicing	1024	167	1110	83	92.252252	10	5
Drying	1395	220	1475	80	94.576271	15	6
Cooling	1462	270	1600	138	91.375	10	5
Packing	1180	130	1310	130	90.076336	10	6
Shipping inventory	0	5800	0	0		0	4
Total	5619	13123				60	

Future PCE of Bread production floor

At future state, VD and NVD time was calculated consecuitively 5619 sec and 13123 sec, so the lead time was 24078sec. As earlier it was mentioned that -
PCE = (Value Added time/Lead time)*100%

PCE = 29.9807918

Present Takt time of Bread production floor

Working shift per day	= 2 shift
One shift time	18742 sec
Two shift time	37484 sec
Total change over time	= 1 hour or 3600 sec
Launch time	= 30 minutes or 1800 sec
Net available time per day	= Two shift time - (Total change over time + Launch time) =37484 sec -5400 sec = 32084 sec
Daily saleable demand	= 300 kg or 3000 bread

Takt time = 32084/3000 sec = 10.69 sec/bread

It seems that at future state, the PCE would be 29.98% which could be globally competitive

5.Discussion:

Non-value added (NVA) activities are those that do not add value to the product or service from the customer's perspective. By identifying and eliminating NVA activities, bakeries can improve the efficiency and accuracy of their inventory management process, which can lead to reduced costs, improved quality, and increased profitability. Bakeries can improve their efficiency, accuracy, and profitability by eliminating NVA activities in the receiving, mixing, slicing, drying, cooling, packaging, and shipping processes. Bakeries can improve their efficiency, accuracy, and profitability by eliminating NVA activities in the receiving, mixing, slicing, drying, cooling, packaging, and shipping processes.

Here is example of some (NVA) activities that must be reduced :

- Receiving – manual data entry, Lack of organization, inefficient communication etc.
- Mixing –Unnecessary handling, Inefficient mixing methods, Lack of organization, Inefficient clean up etc.
- Slicing, Drying, Cooling and Packaging–Lack of automation, Inefficient clean up, Poorly calibrated equipment or setup and waiting time for the next step could be addressed etc.
- Shipping inventory –Waiting times due to transportation, Inefficient layout, Poor scheduling etc.

In addition to the above, bakeries can also identify and eliminate NVA activities by conducting process mapping and value stream mapping exercises. These exercises can help to identify all of the steps involved in the inventory management process and to identify which steps are adding value to the customer and which steps are not. Once NVA activities have been identified, they can be eliminated or streamlined to improve the overall efficiency of the process.

6.Conclusion:

In this study, an attempt has been made to improve the productivity and make the speed up of "SunBest" Bread company by using lean tools. The basic improvements were observed for lead time, PCE, and Takt time between present and future state. At present state, the lead time, PCE and Takt time were observed consecutively 32605 sec, 12.82% and 19.94 sec/bread (200g). However, it was calculated that after the implementation of lean tools at future state, the lead time, PCE and Takt time would be consecutively 18742 sec, 29.98%% and 10.69 sec/ bread (200g). Finally, it was suggested to the bread company to implement the proposed lean tools prior to get the expected short lead time.